3SD-Net: SAR Small Ship Detection Neural Network

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Abstract*— This article studies a practically meaningful ship detection problem from synthetic aperture radar (SAR) images by neural network. We broadly extract different types of SAR image features and raise the intriguing question whether these extracted features are beneficial to: 1) suppress data variations (e.g., complex land-sea backgrounds, scattered noise) of real-world SAR images and 2) enhance the features of ships that are small objects and have different aspect (length–width) ratios, therefore resulting in the improvement of ship detection. To answer this question, we propose an SAR-ship detection neural network (called 3SD-Net for short), by newly developing bidirectional coordinate attention (BCA) and multiresolution feature fusion (MRF) and a center point distribution module (CPDM) based on CenterNet. In detail, we first develop BCA to make 3SD-Net focus on ship features as much as possible while ignoring the background noise. Second, we leverage MRF to enhance the spatial information of small-scale ships yet solve the nontrivial problem of small-scale and shallower pixels easily lost after deep convolution in SAR images. Moreover, considering the varying length–width ratio of arbitrary ships, we study the probability distribution around the ship center. We concentrate on enhancing the distribution function of the ship center, thereby significantly improving the performance of the basic CenterNet detector. This improvement is achieved without incurring additional computational and time costs. The experimental results obtained from the public SAR-Ship and SSDD datasets demonstrate the superior performance of our method compared with its competitors. Specifically, our 3SD-Net achieves average precision (AP) values of 91.66% and 90.22% on the two datasets, respectively, outperforming YOLOV7 (90.31% and 87.32%) and EfficientVit (90.06% and 90.08%). *Source code will be released upon publication.

***Index Terms*— Bidirectional coordinate attention (BCA), central probability distribution, multiresolution feature fusion (MRF), ship detection, synthetic aperture radar (SAR)-ship detection neural network (3SD-Net).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active microwave imaging sensor with long-distance observation capability in all-day and all-weather conditions. Thus, it has good adaptations for monitoring oceans. In SAR images, ships are the most critical yet small targets to detect. SAR-ship detection aims to find the predefined ship objects in a given SAR scene by generating accurate 2-D bounding boxes to locate them. SAR-ship detection is widely used in ship tracking (e.g., maritime transportation), ocean event detection (e.g., maritime rescue), and ocean scene understanding. Although many efforts have been explored for the SAR-ship detection task, it is still not completely and effectively solved, due to the nontrivial SAR imaging mechanism, where various ships are very small and blurred, and even submerged in extremely complicated backgrounds.

Traditional SAR target detection methods are mainly based on contrast information [1], [2], [3], geometric [4] and texture features [5], and statistics [6], [7]. They are implemented by the hand-crafted feature extractors and classifiers, e.g., SIFT operator [8], Harr operator [9], and HOG operator [10]. However, these methods not only are time-consuming but also have inaccurate detection results in complicated sea-and-land scenarios. For example, the constant false alarm rate (CFAR) detectors [11] determine the detection threshold according to a pre-established clutter statistical model. The model is highly dependent on manually designed features and is easily affected by noise in pictures. In summary, there are shortcomings such as low efficiency and weak generalization ability for these traditional detection wisdoms.

Recent years have witnessed the boom of convolutional neural network (CNN)-based detection models, which can automatically extract features and get rid of the shortcomings of manually designed features [12] for SAR-ship detection. These models can be divided into two categories, i.e., the first ones are based on the two-stage architectures such as feature pyramid networks (FPNs) [13], R-CNN, Fast R-CNN [14], and Mask R-CNN, and the second are based on the one-stage architectures such as SSD [15], YOLOs [16], [17], and RetinaNet [18]. Two-stage detectors will divide object detection into two stages: in the first stage, the proposed regions are generated to obtain the approximate proposal and probability of the object, and the second stage will regress the position, size, and category of object bounding boxes to the proposed regions through another network. The core of the one-stage object detectors is to omit the generation of proposed regions for the input image and directly regress the object size, position, and category through the network, which is faster than the two-stage detector. Naturally, precision and efficiency of these deep models are often superior to the traditional methods, once large labeled data are consumed for training the well-designed networks. Although these advanced detectors have achieved great results in many fields of recognition and detection,

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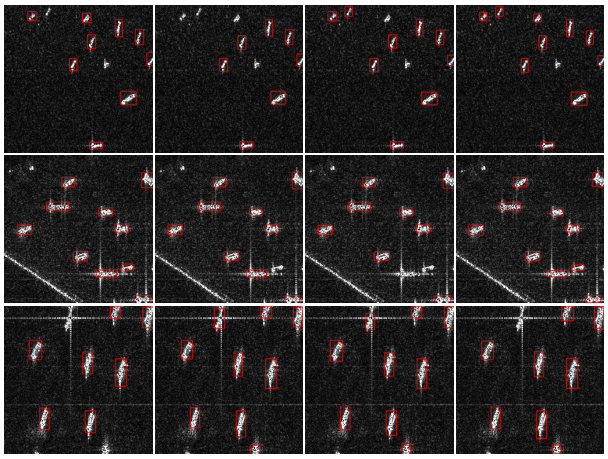


Fig. 1. Ships are often small targets and submerged in highly complex backgrounds, posing significant challenges for accurate SAR-ship detection. In addition, SAR images inherently suffer from speckle noise, further complicating the detection process. Therefore, when designing an effective neural network model, it is crucial to suppress the adverse factors present in the surroundings while enhancing the beneficial features associated with ship targets. Our 3SD-Net can deal with the aforementioned problems, leading to better detection results than SOTAs. (a) CenterNet. (b) YOLOV7. (c) EfficientDet. (d) Ours.

there are still many problems with these detectors in SAR ship detection. These problems include the misclassification caused by the high similarity between ships and islands in complex sea-and-land scenes and the omission of small target ship detection under long-distance satellite observations. Therefore, directly using the detector for SAR-ship detection often fails to achieve good results.

Fig. 1 shows these three types of SAR-ship detection challenges, where the local regions are similar to small ship targets spread over the whole background. Ships falling within a limited pixel range are particularly susceptible to interference from clutter and noise, presenting a challenge for accurate detection with the existing advanced detectors. We can see that CenterNet has a poor detection effect on ships in small targets and complex backgrounds, and many ships are missed. Meanwhile, YOLOV4 and EfficientDet also missed or mistakenly detected ship targets in complex SAR images. Thus, exploring the interaction information amongst SAR image features in large-range dependencies to amplify the difference between the ship target and its background is crucial for robust detection. However, cutting-edge learning models are limited by the locality of CNNs, which behave poorly to capture large-range dependencies.

To solve these challenges, we design a high-speed and effective detector called 3SD-Net. First, we propose a new attention mechanism, i.e., bidirectional coordinated attention (BCA). The detection of ships in SAR imagery is often hindered by the complex background noise and the presence of islands. To circumvent issues related to noise extraction during feature aggregation, we propose a novel attention mechanism BCA. Drawing inspiration from the coordinate attention (CA) mechanism, BCA integrates various pooling methods to enhance the performance. Next, considering the small target ships and shallow pixels caused by long-distance observation, we adopt a multiscale approach to solve the problem. Since SAR images are gray-scale images and only have ship targets, they contain much less feature information than ordinary color optical images. Therefore, we propose a simple multiscale feature fusion module MRF in SAR ship detection. We generate high-resolution feature maps in different feature layers for feature fusion. Through MRF, the detection effect of small targets is improved without significantly increasing the model size, and the entire ship detector network is made simpler and faster. Finally, considering the change in detection effect caused by the aspect ratio of ships, we study the influence of distribution functions around different centers on the detection effect in the field of ship detection. The circular Gaussian probability distribution of CenterNet is augmented with additional distribution functions in our model, which significantly improves the detection effect of the detector without incurring additional computational and time costs.

Our main contributions are as follows.

1) We propose a BCA and multiresolution feature fusion (MRF) network, which we call 3SD-Net.
2) We propose a BCA mechanism for SAR-ship detection. BCA makes the network pay more attention to ship features in complex backgrounds.
3) We propose an MRF module. It achieves good results for small target ship detection, but does not greatly increase the size of the model and reduce the detection speed.
4) We focus on the effect of ships with different aspect ratios and study the effect of different distribution functions around the center on ship detection.

This article is extended from our conference version of SARShipNet [19]. In the conference version, we have shown commendable performance in SAR detection, but the detection of small target ships in SAR imagery is not well-solved in our solution. Small target detection in SAR imagery is a challenging problem, and there is still ample scope for improving the efficacy of our initial work on small target detection. Therefore, we make efforts to improve our initial work to address the small target detection problem more effectively. To be specific, this work extends our initial study from the following aspects.

1) We investigate the utilization of diverse probability distribution models in proximity to the ship center, propose a center point distribution module (CPDM) module, and assess their impact on ship detection using four distinct types of probability distribution models. The experimental findings demonstrate the merits associated with these four distributions.
2) We conduct a comprehensive analysis of the SARShipNet model's performance, using additional ablation experiments and comparative studies to evaluate our ship detection model in SAR images.
3) We provide extensive visualizations of the detection results obtained by our model, highlighting its superior capabilities in detecting small target ships.

II. Related Work

SAR target detection aims to identify and locate target objects on the ground by analyzing the radar data returned. SAR operates by mounting radar equipment on an aircraft,

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such as a satellite, aircraft, or drone, and sending radar pulses to the ground while receiving the signals. These return signals contain reflection information from ground targets. The common methods for SAR target detection can be categorized into one-stage and two-stage methods.

A. One-Stage Detectors

The bounding box and category of the object are directly obtained from the feature map of input images. YOLOV3 [16] divides the image into multiple grids of different sizes, which are used to predict objects of different sizes, achieving a balance between speed and accuracy. SSD [15] uses different sizes of convolutional layers to use anchor boxes with different sizes and numbers for object detection. RetinaNet [18] proposed that focal loss is used to balance the problem of a serious imbalance in the proportion of positive and negative samples. CenterNet [20] and fully convolutional one-stage (FCOS) object detection [21] abandon the setting of the anchor and implement anchor-free object detection. CenterNet regards the object as a point, and FCOS performs pixel-by-pixel object detection. These one-stage detectors achieve a balance of speed and accuracy.

B. Two-Stage Detectors

The region proposal is used to rough regression and classify the regions of the object and the probability of the foreground, and then perform fine classification and regression through another network. The classic two-stage detector: Faster R-CNN [14] uses RPN instead of the region search algorithm to achieve end-to-end training. FPN [13] first performs bottom–up feature extraction and then performs top–down feature fusion to increase the semantic information of all the feature layers. Mask R-CNN [22] is a branch that adds a prediction segmentation mask based on Faster R-CNN. CascadeRCNN [23] achieves the goal of continuously optimizing the prediction results by cascading several detection networks. Two-stage detection achieves very good accuracy in many tasks, but it cannot balance accuracy and detection speed.

C. Ship Detection

In recent years, due to the great application value of ship detection, many researchers have been attracted. The traditional SAR ship detection algorithm, CFAR detectors [24], is one of the most commonly used techniques. Gao et al. [25] considered practical application situations and tried to strike a good balance between estimation accuracy and speed. Leng et al. [26] introduced a bilateral CFAR algorithm for ship detection and reduced the influence of SAR ambiguity and ocean clutter. Shi et al. [27] used a histogram of oriented gradients for feature extraction and carried out the separation of the ship from the background.

With the development of deep learning, many researchers have begun to use deep learning for SAR ship detection. Cui et al. [28] integrated the FPNs with the convolutional block attention module. Du et al. [29] introduced significant information into the network so that the detector could pay more attention to the object region. Fu et al. [30] proposed an anchor-free network for ship detection, using a balancing pyramid composed of attention-guided and using different levels of features to generate appropriate pyramids. Guo et al. [31] improved CenterNet++ and enhanced ship feature through multiscale feature fusion and head enhancement. Zhao et al. [32] combined the RFB module with the attention module and achieved good detection results. BiFA-YOLO [33] proposes a YOLO-based arbitrary-oriented SAR ship detector, addressing challenges in detecting multiscale, arbitrary-oriented, and densely arranged ships in HR SAR images. CP-FCOS [34] optimized the features of the position regression branch in the FCOS network using a category-position (CP) module. This module generates guidance vectors from the features of the classification branch, thereby enhancing the performance of target positioning in complex scenes.

III. Methodology

Motivation: To develop a high-precision ship detector and address the challenges associated with ship detection, it is imperative to suppress the influence of adverse environmental factors on extracted features while simultaneously enhancing the beneficial features inherent in ship targets themselves. By thoroughly considering both the adverse and beneficial features of SAR images containing ships, we broadly extract different types of SAR image features: 1) suppress data variations (e.g., complex land–sea backgrounds, scattered noise) of SAR images and 2) enhance the features of ships that are small objects and have different aspect (length–width) ratios, therefore resulting in the improvement of ship detection. We propose an SAR-ship detection neural network (called 3SD-Net for short). 3SD-Net is composed of three modules, as shown in Fig. 2. The first module is the feature extraction network that a backbone adds the attention mechanism: BCA. The second module is feature fusion: MRF. The third module is the CPDM. Through the above three corresponding solutions, the problems existing in the field of SAR ship detection are effectively solved, and the deep learning detector still achieves satisfactory detection results in the field of ship detection.

A. Network Structure

The network structure of the model is shown in Fig. 2. ResNet-50 serves as the backbone network in our study, which is divided into five stages. By incorporating our BCA attention mechanism at different stages, the model emphasizes the extraction of effective features while disregarding irrelevant noise features. Following feature extraction, we introduce a feature fusion MRF module that leverages deconvolution to generate three high-resolution feature maps from various ResNet-50 layers for enhanced detection of small target ships. Subsequently, we proceed to the detection stage where the entire detection head comprises three branches: classification branch, width–height regression branch, and center point offset branch. In the classification branch, we propose a CPDM module to investigate how probability distribution around

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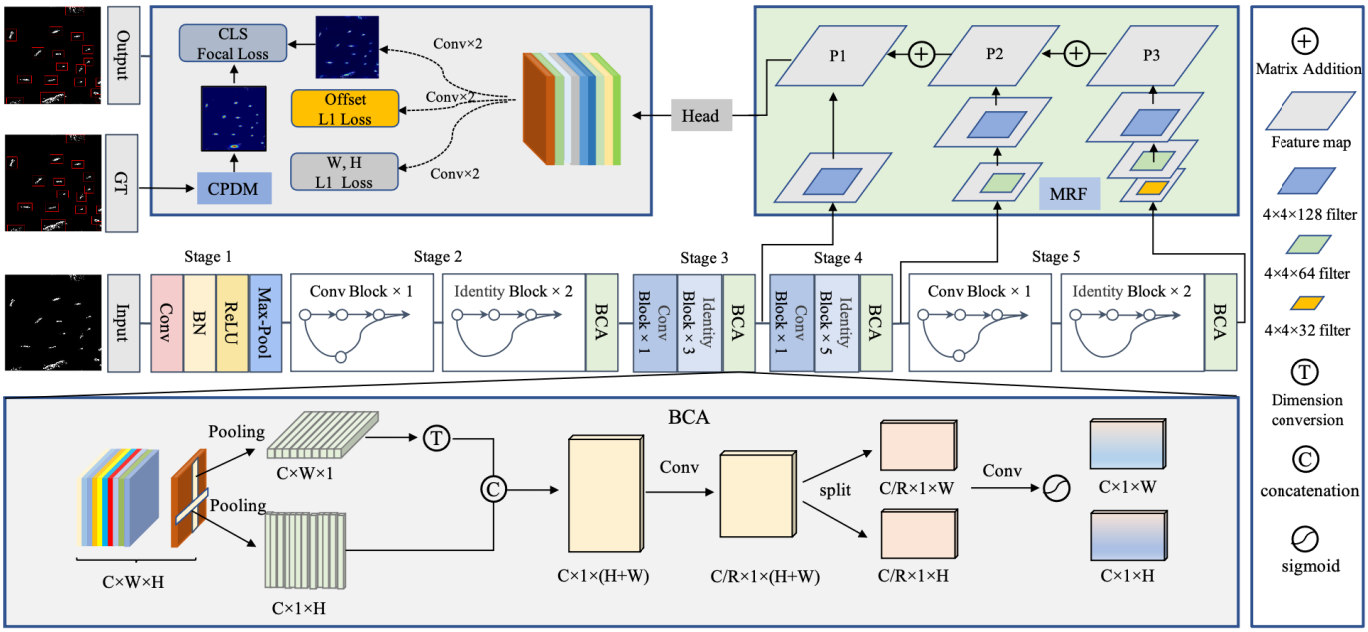


Fig. 2. Overview of our 3SD-Net. 3SD-Net is composed of three modules: the feature extraction network that the backbone adds the attention mechanism, feature fusion: MRF, and CPDM.

different ship centers influences detection performance. After the inspection, the ship will be boxed out according to the original drawing.

B. Bidirectional Coordinate Attention

Due to the characteristics of complicated background islands along with other scattered noise, the effectiveness of the ship detection models is severely reduced, which makes it difficult to accurately detect ships. The primary objective of the BCA attention module is to address this issue by enhancing ship detection accuracy amidst complex backgrounds. Inspired by the CA mechanism [35], we propose a BCA mechanism. CA mechanism decomposes channel attention into two 1-D feature encoding processes, which aggregate features along with two spatial directions, respectively. In this way, remote dependencies can be captured in one spatial direction, while accurate location information can be retained in another spatial direction. But there is a lot of noise redundancy in SAR pictures, such as speckle noise or complex land background. It is indisputable that the use of a CA mechanism can enhance the efficacy of feature weight extraction. However, given the coexistence of SAR images containing numerous speckled and scattered noises alongside ships, there is a possibility for noise features close to ships to also be extracted. While disregarding redundant noise information in SAR images is essential, relying solely on average pooling for information aggregation inevitably results in the extraction of noisy features. In contrast to the CA mechanism, we propose that both average and max pooling are efficient methods for information aggregation. Thus, we propose a BCA mechanism that combines average pooling and max pooling, as shown in Fig. 3.

BCA is formulated as

$$\begin{aligned} f_a &= \delta\left(F1\left[\text{avgpool}\left(x_c^h\right), \text{avgpool}\left(x_c^w\right)\right]\right) \\ g^h, g^w &= \sigma\left(F_h\left(f_a^h\right)\right), \sigma\left(F_w\left(f_a^w\right)\right) \\ y_c(i,j) &= x_c(i,j) \times g_c^h(i) \times g_c^w(j) \end{aligned} \tag{1}$$

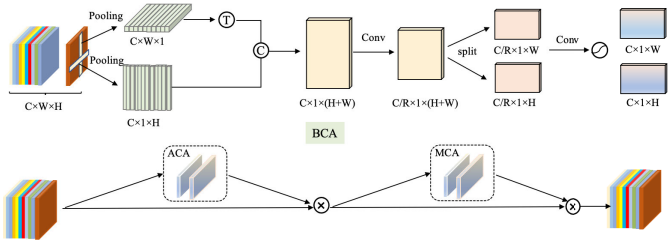


Fig. 3. BCA mechanism. Adaptive channel attention (ACA) uses avgpooling to aggregate features, and multi-scale channel attention (MCA) uses maxpooling to aggregate features.

$$\begin{aligned} f_m &= \delta\left(F_2\left[\text{maxpool}\left(y_c^h\right), \text{maxpool}\left(y_c^w\right)\right]\right) \\ z^h, z^w &= \sigma\left(F_{h2}\left(f_m^h\right)\right), \sigma\left(F_{w2}\left(f_m^w\right)\right) \\ \text{output}(x_c(i,j)) &= x_c(i,j) \times g_c^h(i) \times g_c^w(j) \\ &\quad \times z_c^h(i) \times z_c^w(j) \end{aligned} \tag{2}$$

where $x \in R^{C\times W\times H}$ is the feature map, c represents the channel index, and $\text{avgpool}(x_c^h)$ and $\text{avgpool}(x_c^w)$ represent the average pooled output of the cth channel with height h in the horizontal direction and width w in the vertical direction, respectively. [] represents the splicing operation of the feature map. $F1$ represents the 1×1 convolution. δ is the nonlinear activation function, and $f_a \in R^{(C/r)\times(W+H)\times 1}$ is the intermediate feature. $f_a^h \in R^{(C/r)\times H\times 1}$ and $f_a^w \in R^{(C/r)\times W\times 1}$ are two vectors obtained by decomposing f_a, F_h, and F_w are two 1×1 convolutions. σ is the sigmoid activation function. $g^h \in R^{C\times H\times 1}$ and $g^w \in R^{C\times W\times 1}$ are two attention weights, respectively. $y_c(i,j)$ is the feature point output after avgpool attention. Similarly, the process of using the maxpool attention mechanism is consistent with the avgpool attention mechanism. $\text{Output}(x_c(i,j))$ is the last output of attention through BCA. BCA makes full use of the captured position information through two different information aggregation methods so that the region of interest can be accurately captured.

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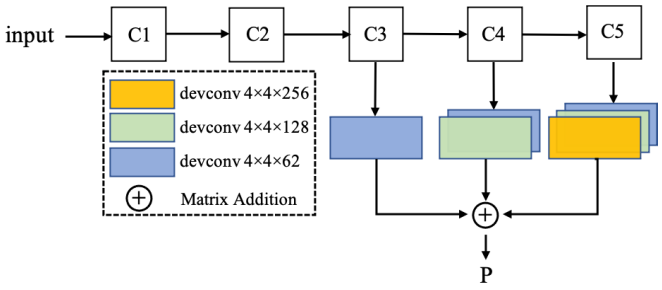


Fig. 4. MRF module. C3, C4, C5 are the feature layers output by the last three stages of ResNet50, Devconv is the deconvolution operation.

We take ResNet-50 as the backbone and divide it into five stages according to the size of the feature maps of different layers. In the bottom–up path, the output of the last five stages is defined as {C1, C2, C3, C4, C5}. As shown in Fig. 2, {C1, C2, C3, C4} add BCA mechanism in the middle, and this process is summarized as

$$C_i = \begin{cases} \mathrm{BCA}(C_i), & \text{if } i = 2, 3, 4, 5 \\ C_i, & \text{else } i = 1. \end{cases} \tag{3}$$

C. Multiresolution Feature Fusion

The problem of small target ships and shallow pixels caused by long-distance observation seriously affects the effect of ship detectors. Considering that the feature information contained in SAR images is far less than that of ordinary color optical images, we propose a straightforward multiscale method MRF. In contrast to other methods that solely involve basic feature fusion processing, our approach incorporates the distinctive characteristics of SAR images by generating high-resolution feature maps across multiple layers, thereby facilitating comprehensive feature fusion. The MRF method solves the problem of small targets without significantly increasing model size and makes the whole detection pipeline very simple.

The output of the last three stages of ResNet-50 is defined as {C3, C4, C5}. The MRF module uses three feature layers to generate three feature maps of the same size. Fig. 4 shows the implementation details of the MRF module. Through multiple rounds of deconvolution, the high-resolution feature maps of {C3, C4, C5} are ultimately merged to enhance the detailed features of the ship. The process can be defined as

$$p_i^j = \mathrm{Dconv}\left(C_j, i, S_i\right), \quad i \in \{1, 2, 3\}, \quad j \in \{3, 4, 5\} \tag{4}$$

$$P = \mathrm{ReLU}\left(p_1^3 \oplus p_2^4 \oplus p_3^5\right) \tag{5}$$

where Deconv is the deconvolution [36] operation, i is deconvolution times, C_j is the jth layer feature map of ResNet50, S_i is the size set of the target feature map obtained after deconvolution, where $S \in \{[128], [64, 128], [32, 64, 128]\}$, and p_i^j represents the feature map obtained by i times deconvolution of the jth layer feature of ResNet50. $\oplus$ denotes elementwise addition, and rectified linear unit (ReLU) is the activation function. P is the total feature after fusion.

D. Center Point Distribution Module

The probability distribution function centered around the object's centroid plays a crucial role in training with the CenterNet network. To facilitate supervised training, we use a distribution function to map the ground-truth (GT) center point onto the heat map. This distribution function aims to enhance tolerance toward predicted offsets from the center point, allowing predictions near the true center to be considered acceptable. By treating objects as points, CenterNet achieves remarkable detection performance. However, previous studies commonly adopted CenterNet's circular Gaussian probability distribution directly as their chosen distribution function.

When directly applying the distribution function of CenterNet to the field of ship detection, we have observed limitations, particularly when ship targets exhibit large aspect ratios. To address this issue, we investigate the impact of the distribution function around different centers and propose the CPDM module to alleviate the influence due to large aspect ratios. We design four distribution functions, including circular Gaussian distribution (CGS), elliptical Gaussian distribution (EGS), exponential distribution (EDS), and logarithmic distribution (LDS), to study the performance of the detector under different distribution functions.

For each GT, the key point $p \in R^2$ corresponding to category c, then calculate the key points after downsampling $\tilde{p} = \lfloor p/R \rfloor$. CenterNet splats all the GT keypoints onto a heatmap $Y \in [0, 1]^{(W/R)\times(H/R)\times C}$ using distribution functions.

CGS is formulated as

$$\mathbf{Y}_{x,y,c} = \exp\left(-\frac{\left(x - \tilde{p}_x\right)^2 + \left(y - \tilde{p}_y\right)^2}{2\sigma_p^2}\right) \tag{6}$$

where σ_p is an object size-adaptive standard deviation.

EGS is formulated as follows:

$$\mathbf{Y}_{x,y,c} = \exp\left(-\frac{\left(x - \tilde{p}_x\right)^2}{2\sigma_x^2} - \frac{\left(y - \tilde{p}_y\right)^2}{2\sigma_y^2}\right) \tag{7}$$

where $\sigma_x = (\alpha w/6)$, $\sigma_y = (\alpha h/6)$, α is a superparameter, and w and h are the width and height of GT, respectively.

EDS is formulated as

$$\mathbf{Y}_{x,y,c} = e^{-\left(\left|x-\tilde{p}_x\right| + \left|y-\tilde{p}_y\right|\right)}. \tag{8}$$

LDS is formulated as

$$\mathbf{Y}_{x,y,c} = \frac{1}{1 + \log_{10}\left(\left|x - \tilde{p}_x\right| + \left|y - \tilde{p}_y\right| + 1\right)}. \tag{9}$$

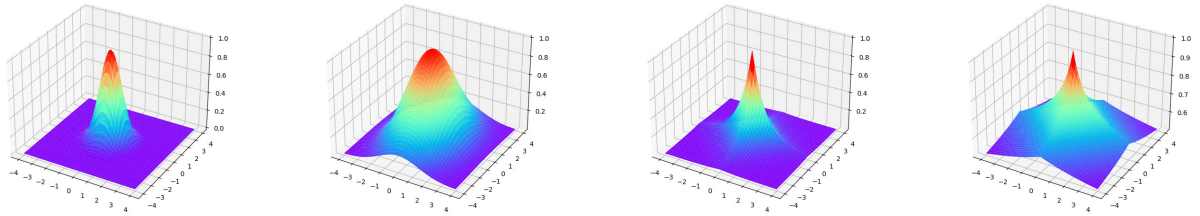
Fig. 5 shows the distribution of different distribution functions in the 3-D coordinate system. Fig. 6 shows the probability distribution around the center point of the ship under different distribution functions through a heatmap. The visualization of these distributions can help us understand and analyze CPDM modules more clearly.

E. Loss Function

The features after feature fusion are used for classification and regression prediction. The head of our model has three branches: the classification, the bounding box regression, and the center point regression, so the loss function is defined as

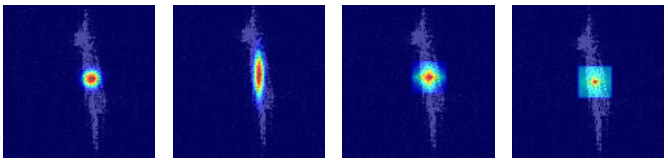
$$L = L_{\mathrm{cls}} + \lambda_1 L_{wh} + \lambda_2 L_{\mathrm{point}} \tag{10}$$

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(a) (b) (c) (d)

Fig. 5. 3-D coordinates of different probability distribution functions are visualized. The z-axis represents the size of the activation value. The activation value of the center point is equal to 1, and its surrounding activation values show a decreasing trend as the distance increases. (a) CGS. (b) EGS. (c) EDS. (d) LDS.



(a) (b) (c) (d)

Fig. 6. Visualization of the 2-D heatmap distribution of different probability distribution functions around the target center point. The activation value at the center of the object is 1, and the activation value becomes smaller as it moves away from the center point. (a) CGS. (b) EGS. (c) EDS. (d) LDS.

where L_{cls} is the classification loss, L_{wh} is the bounding box regression loss, and L_{point} is the center point regression loss. λ_1 and λ_2 are the weight parameters. We set $\lambda_1 = 0.1$ and $\lambda_2 = 1$, respectively. L_{cls} is the focal loss formulated as

$$L_{\text{cls}} = \frac{-1}{N}\sum_{xyc}\begin{cases}(1-\hat{y})^{\alpha}\log(\hat{y}), & \text{if } y=1\\ (1-y)^{\beta}(\hat{y})^{\alpha}\log(1-\hat{y}), & \text{otherwise}\end{cases} \tag{11}$$

where $\hat{y}$ is the confidence of classification prediction, y is the GT category label, α and β are the hyperparameters in focal loss, where $\alpha = 2$ and $\beta = 4$ (similarly reported in [20]), and N is the number of positive samples.

L_{wh} is the bounding box regression loss using the $L1$ loss function. L_{wh} can be defined as

$$L_{wh} = \frac{1}{N}\sum_{i=1}^{N}\sqrt{\left(\hat{w}_i - w_i\right)^2 + \left(\hat{h}_i - h_i\right)^2} \tag{12}$$

where $\hat{w}_i$ and $\hat{h}_i$ are the width and height of the predicted bounding box, respectively, and w_i and h_i are the width and height of the GT bounding box, respectively.

Besides, the size of the SAR original image and feature image is not aligned due to downsampling. To address this problem, the center point offset ΔS is taken into account. L_{point} is the center point regression loss using the $L1$ loss function. L_{point} can be defined as

$$L_{\text{point}} = \frac{1}{N}\sum_{i=1}^{N}\left|\Delta\hat{S} - \left(\frac{S}{R} - \left\lfloor\frac{S}{R}\right\rfloor\right)\right|. \tag{13}$$

S is the bounding box center coordinate, and R is the downsampling multiple, where $R = 4$. $\lfloor S/R \rfloor$ represents the downsampling of the center of the object, and the decimal part is ignored. $(S/R) - \lfloor S/R \rfloor$ represents the accuracy loss during downsampling. $\Delta\hat{S}$ is the predicted center point $(\hat{x}_i, \hat{y}_i)$ offset $(\sigma\hat{x}_i, \sigma\hat{y}_i)$. Finally, the coordinates of the upper left and lower right corners of the predicted bounding box are defined as

$$\left(\hat{x}_i + \sigma\hat{x}_i - \frac{\hat{w}_i}{2}, \hat{y}_i + \sigma\hat{y}_i + \frac{\hat{h}}{2}, \hat{x}_i + \sigma\hat{x}_i + \frac{\hat{w}_i}{2}, \hat{y}_i + \sigma\hat{y}_i - \frac{\hat{h}}{2}\right). \tag{14}$$

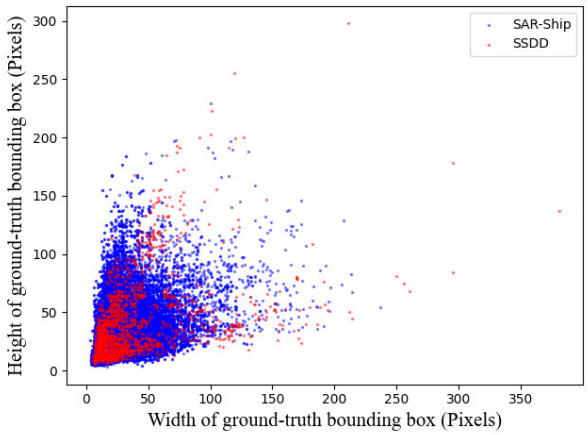


Fig. 7. Distribution of ship GT under the SAR-Ship and SSDD datasets. From the scattered points, it can be seen that there are a large number of small-sized ship targets, and the ships have different aspect ratios. This is very different from ordinary target detection.

IV. Experiments

A. Datasets and Experimental Settings

1) Datasets: We directly evaluate the 3SD-Net model on the SAR-Ship [37] and SSDD [38] datasets. The SAR-Ship dataset is constructed with 102 domestic GF-3 SAR images and 108 Sentinel-1 images. The original data covers multiple resolutions and different polarization modes of 3, 5, 8, 10, and 25 m, and it contains 43 819 ship slices of 256 pixels × 256 pixels, totaling 59 535 ship samples. The SSDD dataset contains 1160 images and 2456 ships, with an average of 2.12 ships per image. These images cover imaging results under a variety of conditions, making the performance of the detector more robust. Ships in SSDD are generally small, with an average size of only about 35 × 35 pixels, which is very difficult for traditional detection algorithms. Fig. 7 shows the distribution of ship sizes and ratios in datasets. In our research, small ships are defined according to the reference information

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TABLE I
EXPERIMENTS WITH DIFFERENT DISTRIBUTION FUNCTIONS

Distribution function	Precision	Recall	F1	AP0.5
CenterNet+CGS	94.73	60.28	74	88.32
CenterNet+EGS	93.60	71.71	81	**88.97**
CenterNet+EDS	**95.81**	59.87	73	87.54
CenterNet+LDS	79.18	**85.95**	**82**	87.26

provided in the SAR-Ship dataset. Specifically, we define M as the number of pixels in an image. Subsequently, a ship is defined as a small target if its label box area is less than $M \times 1.04\%$. We divide the dataset into the training set and test set at a ratio of 8:2.

2) Implementation Details: The experiment is implemented based on the PyTorch framework. To train the model, our deep learning environment is deployed on the Ubuntu18.04 system. The whole training framework environment adopts the combination of (CUDA)v11 + cudnn v8.0 + pytorch1.7. We use ResNet-50 as the backbone of our 3SD-Net and CenterNet as the baseline. All the experiments are performed on RTX3080 with 10 GB of memory. Our backbone is randomly initialized for training without loading the pretraining model. In the training process, the input image size is set to 512 × 512. The optimizer uses Adam [39] algorithm. To ensure a stable reduction of training loss during training, the initial learning rate for the first 20 epochs is 0.001, and the initial learning rate for every five epochs is 0.98 times the original learning rate. The whole training process is 100 epochs, and the batch size is set to 8. For the fairness of model comparison, the value of intersection over union (IOU) is set to 0.5 during evaluation.

3) Evaluation Criteria: To evaluate the detection performance of our model and its competitors, we use four criteria: recall (R), precision (P), $F1$, and mean average precision (mAP) to evaluate their performance. Their formulas are

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{15}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{16}$$

$$F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{17}$$

$$\text{mAP} = \int_0^1 P(R) dR \tag{18}$$

where true positive (TP) is the number of correct ship detection, false positive (FP) is the number of ships that have false alarms, and false negative (FN) is the number of ships that have missed detection.

B. Distribution Function Analysis

The CPDM module is used to study the influence of different distribution functions on the ship detection effect, which may be the first research on the design of distribution functions for ship target detection scenarios. We design four distribution functions, including CGS, EGS, EDS, and LDS. We set the experimental dataset as SAR-Ship. Table I shows the experimental results of different distribution functions on the CenterNet. It can be seen that the EGS distribution achieves the best average precision (AP) (88.15), the EDS has the best precision (95.76), and the LDS has an extremely high recall (85.90). By considering the aspect ratio of the ship, EGS distribution is more suitable for the shape of the ship, and as shown in Fig. 6(b), it achieves the best comprehensive performance. The EDS distribution makes the activations farther away from the center drop faster, as shown in Fig. 5(c), so that the larger activations are concentrated near the center to achieve the best precision. LDS allows points farther from the center point to have larger activation values, as shown in Fig. 5(d), which makes it possible for a target to have multiple center points with larger activation values at the same time, and the predicted center points increase, thus achieving the best recall rate, but precision is very low. AP is a small target evaluation metric.

TABLE II
3SD-Net TEST RESULTS UNDER DIFFERENT α CONDITIONS

Parameter	Precision	Recall	F1	AP0.5
α = 0.1	96.31	6.19	12	88.09
α = 0.2	97.93	7.34	14	88.72
α = 0.3	**98.42**	24.41	39	89.34
α = 0.4	97.77	42.13	59	90.01
α = 0.5	96.05	63.27	76	90.07
α = 0.6	97.64	55.93	71	91.22
α = 0.7	96.87	61.72	75	90.96
α = 0.8	95.14	**71.4**	**82**	**91.71**
α = 0.9	96.17	65.25	78	91.48
baseline(CGS)	97.26	50.31	66	90.15

The EGS distribution achieves the best AP, and we search for the hyperparameter α. Table II shows the experimental results combining the three modules of BCA and MRF and EGS or CGS under the SAR-Ship dataset. It can be seen that using a larger α can achieve a better AP value. Compared with CGS, the EGS distribution increases the AP by 0.8 (90.20 versus 89.40) by considering the ship's aspect ratio. The experimental results of these experiments show the importance of the distribution function. We can achieve no-cost performance improvement by changing the distribution function for different detection scenarios. For example, if you want a high recall rate, the LDS distribution may be more appropriate, considering the high-precision EDS is a good choice and considering the overall performance EGS distribution is more appropriate.

C. EGS Distribution Function Analysis

In our proposed method, elliptic Gauss is used to replace circular Gauss, to pay more attention to the change in detection effect caused by ship aspect ratio, in which the superparameter α directly affects the detection effect of the detector. As shown in Table II, we have conducted several experiments with different α, and the performance is unstable. It can be observed that the 3SD-Net of different α is quite different from the CenterNet baseline. When α is small, the obtained map is much lower than the baseline method. With the increase in α, the map is significantly improved, but when α reaches 0.6, it will be unstable, which also shows that there are great differences in the detection effect of different α for the 3SD-Net model. When α is small, 3SD-Net achieves high precision, but the recall is very low. This may be because the radius

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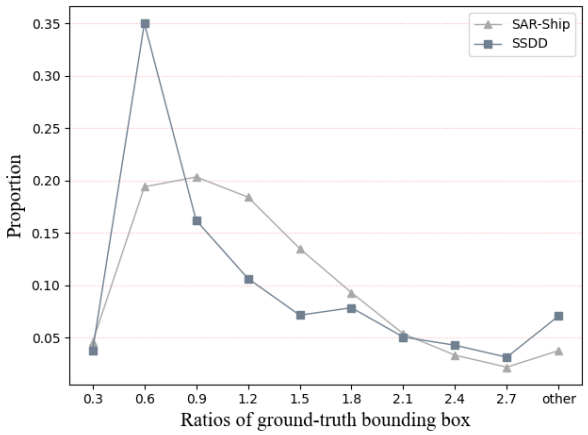


Fig. 8. Ship aspect ratio distribution under the SSDD and SAR-Ship datasets.

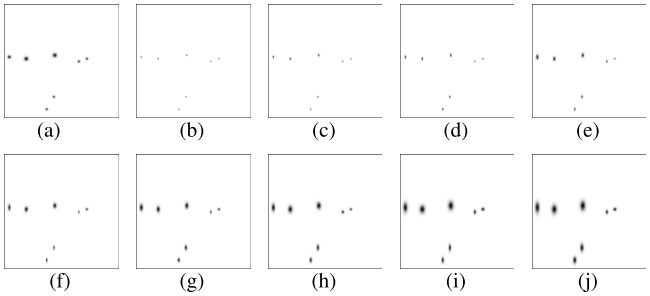


Fig. 9. (a) Heatmap distribution of GT under circular Gauss, and the others are the heatmap distribution of GT under different α. (a) Circular Gaussian. (b) $\alpha = 0.1$. (c) $\alpha = 0.2$. (d) $\alpha = 0.3$. (e) $\alpha = 0.4$. (f) $\alpha = 0.5$. (g) $\alpha = 0.6$. (h) $\alpha = 0.7$. (i) $\alpha = 0.8$. (j) $\alpha = 0.9$.

of Gaussian probability distribution near the center point of the ship is shortened, so only the probability near the center point can be supervised and trained, so that the center point predicted by the model is only close to the GT center point, so many center point predictions far away are missed. Fig. 8 shows the distribution of GT aspect ratio under the SSDD and SAR-Ship datasets. It can be seen that a large number of ships have a large aspect ratio. The attribute of the huge aspect ratio is a huge difference between ship detection and other target detection fields, which should be taken into account. The EGS probability distribution dynamically adjusts the heatmap probability distribution according to different GT aspect ratios, which is more suitable for the scene of ship detection. Therefore, it is necessary to fully consider the impact of the ship's aspect ratio on the detector.

Table II shows the experimental results under different superparameters α. We want to know the EGS of the center of the ship under different α, so we visualize the EGS distribution of the center of the ship. Fig. 9 shows the visualization of heatmap under different α sizes. It can be seen that when α is less than 0.6, the obtained heatmap activation area is much smaller than the baseline circular Gaussian activation area. When α is greater than 0.6, the obtained activation area is larger, and the aspect ratio of the ship is fully considered. According to the above analysis, different α have a great influence on the experimental results, and the experimental results show that a larger α is beneficial to the model. Especially when $\alpha = 0.8$, we get the best mAP (91.71), which

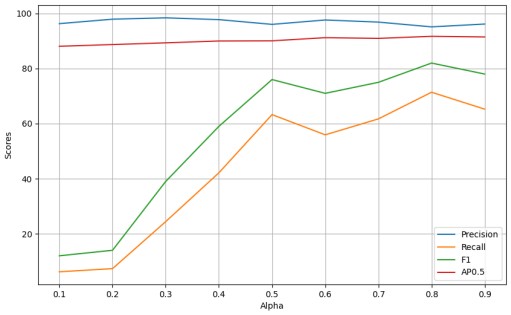


Fig. 10. Line chart of evaluation index change under different parameter α. It can be seen that with the increase in parameter α, recall and $F1$ are significantly improved, and AP0.5 is slightly improved.

TABLE III
RESULTS OF ABLATION EXPERIMENTS

CenterNet CA	CenterNet BCA	MRF	CPDM	Precision	Recall	F1	AP0.5
×	×	×	×	94.83	60.32	74	87.82
√	×	×	×	97.21	51.89	67	89.46
×	√	×	×	96.97	57.83	72	90.04
×	×	√	×	96.28	51.76	67	88.47
×	×	×	√	93.69	**71.84**	82	88.98
×	√	√	×	**97.28**	50.27	66	90.61
×	√	√	√	95.16	71.56	**82**	**91.42**

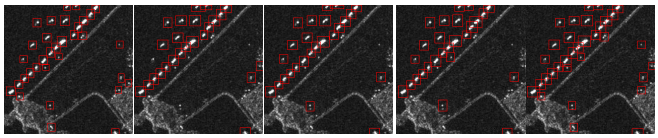


Fig. 11. Detection results of SARShipNet after BCA and MRF are added. (a) GT. (b) Baseline. (c) +CA. (d) +BCA. (e) +BCA + MRF.

has a performance improvement of 1.56% compared with the baseline. Fig. 10 shows the broken line diagram of AP, $F1$, precision, and recall under different parameters α. With the increase in α, it can be seen that precision begins to decline and recall and $F1$ rise, which indicates that the increase in α makes the distance deviation within the large tolerance range of the center point prediction, and the increase in the number of center point predictions leads to the decline of precision and the rise of recall.

D. Ablation Experiment

Here, we verify the effectiveness of different modules (the BCA module, the MRF module, and the CPDM module) in our proposed network on the SAR-Ship large datasets. The results are summarized in Table III. Fig. 11 shows the detection result after each module has been added. The baseline model used in the ablation experiment is CenterNet.

1) BCA Module: Compared with the baseline, the BCA module improves the AP by 2.22%. Compared with CA, BCA achieves a performance improvement of AP 0.58% by adding a max-pooling path for ignoring noise features in SAR images, which also shows the importance of max-pooling information aggregation. From precision and recall in Table III, we observe that the use of a bidirectional attention mechanism improved recall by 11% with a slight decrease in precision compared

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with CA. However, the increased recall demonstrates the sensitivity of BCA module toward detecting ship targets. In particular, we noted that no matter adding CA attention or BCA attention mechanism, the precision of 3SD-Net is improved by at least 2% points, which means that our model classifies noise targets into ships less, and the whole network pays more attention to ships ship feature detection, so it is very useful to add a BCA attention mechanism to the complex SAR ship detection task.

2) MRF Module: From precision and recall in Table III, it can be seen that the MRF module results in both decreases compared with the BCA module, while the highest precision when combined with the BCA module. This means that while the MRF module leads to a minor reduction in semantic information, the combination enhances the detection effect. Only using the MRF module can increase the baseline AP by 0.65%, if combined with the BCA module, it can increase the AP by 0.57% compared with the BCA module. This also shows the fact that better feature fusion can only be done by extracting key features from complex SAR images. Fig. 11(e) shows the detection effect after adding BCA and MRF. It can be seen that whether it is the baseline or adding the CA or BCA module, the detection of small target ships is still difficult. By adding MRF, a simple multiscale method, the size of the model is increased by less than 10 MB. The majority of small target ships were detected.

3) Center Point Distribution Module: Through the analysis of the distribution function experiment, we adopt the EGS as the distribution function of 3SD-Net. Compared with the baseline, using CPDM alone improves AP by 1.16% without adding any cost. If BCA is combined with MRF and CPDM, an AP improvement of 1.38% can be achieved. This also proves that it is very important to choose a distribution function suitable for the task scenario. The EGS can take into account the special attribute of the ship's aspect ratio, which can improve the detection effect of the ship.

4) Quantitative Comparisons: Table III shows the ablation results of the 3SD-Net methods in the SAR-Ship dataset. From AP0.5 in the table, it can be seen that the BCA attention module, MRF module, and CPDM module can improve the detection performance of the baseline model by 2.22%, 0.57%, and 1.16%, respectively. By combining BCA attention with MRF module, baseline AP0.5 can be improved by 2.79%. Compared with only adding BCA attention mechanism, MRF can improve AP by 2.14% on the basis of BCA. CPDM module improves the detection effect by changing the distribution function around the center of the ship by considering the aspect ratio of the ship GT. By adding CPDM module on the basis of BCA and MRF module, AP0.5 is increased by 0.81%. By combining these three modules, our 3S-Net SAR ship detector has achieved the highest 91.42% AP, which is 3.6% AP higher than the baseline (87.82%), which shows the effectiveness of our model.

E. Comparison With Baseline

To be able to compare the performance of detectors fairly, we conduct experiments on the SAR-Ship dataset. The experimental results are shown in Table IV.

TABLE IV
RESULTS OF 3SD-Net AND ITS BASELINES

Dataset	Method	Backbone	AP0.5	APs
SAR-Ship	CenterNet	ResNet-50	88.65	51.62
	3SD-Net	ResNet-50	**91.37**	**53.79**
SSDD	CenterNet	ResNet-50	79.36	48.53
	3SD-Net	ResNet-50	**89.72**	**52.41**

TABLE V
EXPERIMENTS COMPARING BCA WITH SOTA METHODS

Dataset	Method	Backbone	AP0.5	APs
SAR-Ship	CA	ResNet-50	86.79	48.39
	ECA	ResNet-50	87.64	49.66
	CBAM	ResNet-50	88.31	50.13
	BCA(Ours)	ResNet-50	**89.52**	**52.49**
SSDD	CA	ResNet-50	86.24	47.71
	ECA	ResNet-50	86.95	47.39
	CBAM	ResNet-50	87.29	49.61
	BCA(Ours)	ResNet-50	**88.34**	**50.85**

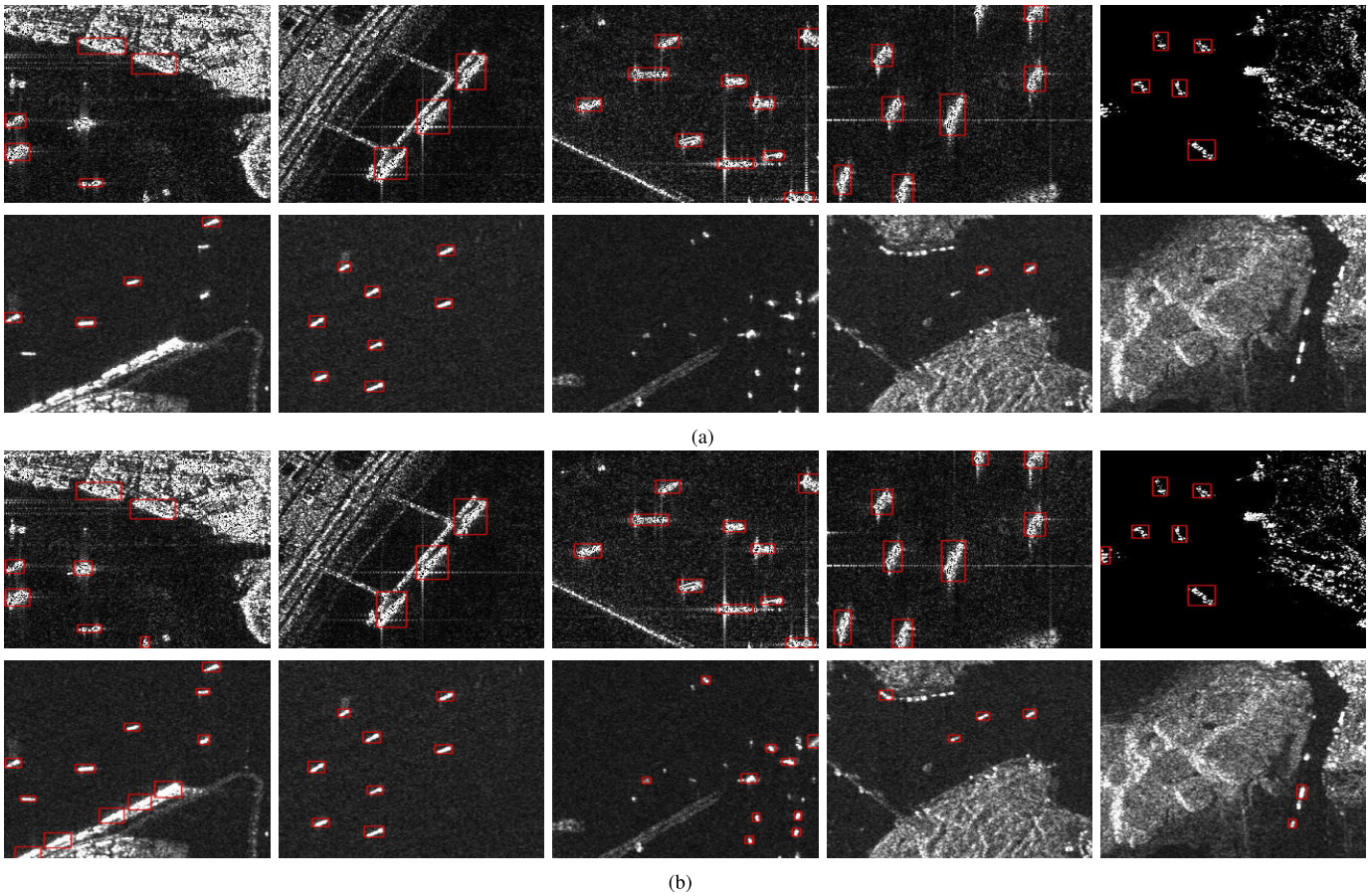
TABLE VI
COMPARISONS OF MRF WITH ADVANCED MULTISCALE FEATURE FUSION METHODS

Dataset	Method	Backbone	AP0.5	APs
SAR-Ship	FPN	ResNet-50	86.74	49.83
	BiFPN	ResNet-50	85.67	47.59
	CFP	ResNet-50	86.32	50.77
	MRF(Ours)	ResNet-50	**88.16**	**51.92**
SSDD	FPN	ResNet-50	85.60	47.39
	BiFPN	ResNet-50	86.76	45.23
	CFP	ResNet-50	86.47	46.73
	MRF(Ours)	ResNet-50	**87.25**	**49.64**

1) Quantitative Comparisons: The 3SD-Net model we proposed has a better performance in evaluation criteria than the CenterNet baseline. This is attributed to the BCA mechanism that makes the network pay more attention to ship features in complex backgrounds, and it is also attributed to the MRF module that enhances the feature information of smaller ships. Elliptical Gauss makes the Gaussian probability distribution of ship center point more consistent with the central distribution of ships with different aspect ratios. 3SD-Net has 91.37% AP and 53.79% APs (Table IV). Compared with CenterNet, the AP and APs achieved by 3SD-Net increase by 2.72% and 2.17%, respectively. Table V shows the comparison between BCA and advanced attention mechanism methods. Different from these methods, BCA performs bidirectional corpus analysis on small targets, so that the model can easily capture small target features. The experimental results show that BCA achieves the best results in the detection of small targets. Table VI shows the comparison between MRF and advanced multiscale feature fusion methods. MRF directly fuses feature maps of three different scales, which well retains the original semantic information of small targets. The results show that MRF outperforms the other methods by an average of 1%–2% points on the APs' metric.

2) Visual Comparisons: Fig. 12 shows the visual comparison results of 3SD-Net and CenterNet under small dataset SSDD. Our 3SD-Net performs better at small object detection.

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(a)

(b)

Fig. 12. Visual comparisons with CenterNet. Inputs are from the test set part of SSDD. Clearly, our 3SD-Net has great advantages in small target detection. (a) CenterNet. (b) 3SD-Net (ours).

F. Comparison With SOTAs

To demonstrate the modules' effectiveness, we compare our proposed modules with other similar modules. We compare the detection ability of CA, ECA [46], CBAM, and BCA on small target ships. The experimental results are shown in Table V. BCA leads the other methods comprehensively on AP0.5 and APs. Among them, BCA outperforms the other methods by an average of 2%–3% points on APs. The above experiments prove that our proposed bidirectional channel attention method is effective in detecting small target ships. We compare the effectiveness of FPN, BiFPN, CFP [47], and MRF on feature fusion. The experimental results are shown in Table VI. The experimental results show that MRF outperforms other feature fusion methods on AP0.5 and APs, especially on APs, MRF is 2.3%–5% points higher than other methods on average. It indicates that MRF can significantly enhance the semantic information of small targets.

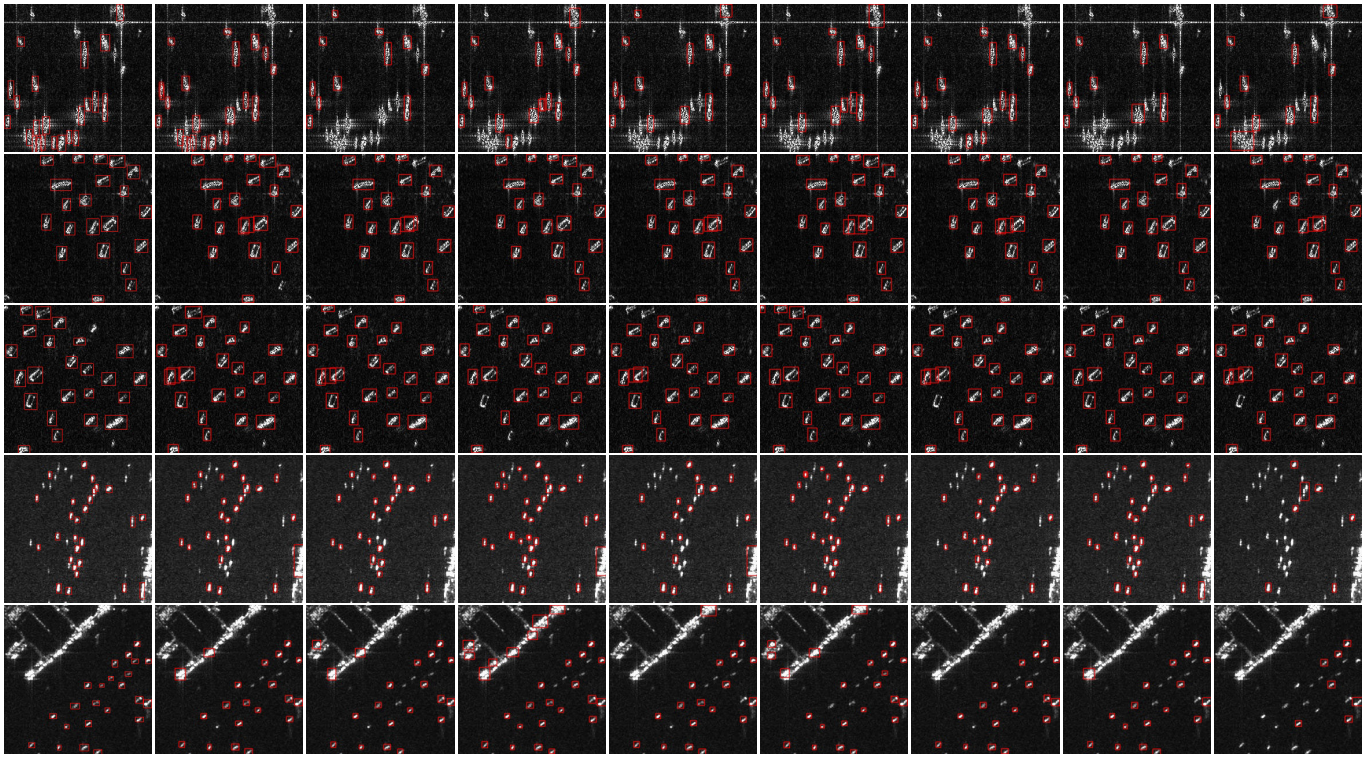
We compare 3SD-Net against a comprehensive set of one-stage and two-stage detectors, including EfficientDet [41], YOLOV4 [17], YOLOX [40], YOLOV3 [16], CenterNet [20], SSD [15], RetinaNet [18], Faster-RCNN [14], YOLOV7 [42], EfficientViT [43], TransNeXt [44], and FocalNets [45]. YOLOv7 excels in real-time scenarios but struggles with complex backgrounds. EfficientViT uses cascaded group attention to reduce computational costs but faces challenges in performance optimization. TransNeXt uses aggregated pixel-focused attention and convolutional GLU to mitigate depth degradation, yet its hybrid architecture incurs high computational costs on large datasets. FocalNets substitute self-attention with focal modulation for enhanced performance, but its generalization capabilities on complex tasks are limited.

1) Quantitative Comparisons: We evaluate our proposed method on five evaluation metrics and compare it with other methods. Table VII shows the quantitative results of all the methods in two datasets. The best result is highlighted in black and the second best results are highlighted in blue. From the table, we can see that our 3SD-Net (*EGS*) model achieves the best AP0.5 value on both the SAR-Ship and SSDD datasets, indicating that our model has the best overall performance. In particular, our model achieves the best $F1$ on the SSDD dataset. Experiments show that our model can achieve the best comprehensive performance on both the SAR-Ship and SSDD datasets and verify the effectiveness of the model. In comparison to EfficientDet, our model increases precision by 5.34%, $F1$ by 2%, AP0.5 by 5.31%, and APs by 4.54% in the SAR-Ship dataset. Compared with YOLOV4, our model increases precision by 0.37%, recall by 1.01%, $F1$ by 1%, AP0.5 by 1.73%, and APs by 2.52% in the SAR-Ship dataset. In comparison to YOLOV7, our model increases precision by 4.53%, $F1$ by 2%, AP0.5 by 1.6%, and APs by 2.04% in the SAR-Ship dataset. When contrasted with EfficientViT, our model increases precision by 6.86%, AP0.5 by 0.14%,

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TABLE VII
EXPERIMENTAL RESULTS OF 3SD-Net AND OTHER DIFFERENT SAR SHIP DETECTORS

Method	Backbone	SAR-Ship					SSDD					FPS	Parameter	Input-size
		Precision	Recall	F1	AP0.5	APs	Precision	Recall	F1	AP0.5	APs			
YOLOV3 [16]	DarkNet53	92.71	70.20	80	88.45	52.71	90.78	67.75	78	80.73	49.30	83	234MB	416×416
YOLOV4 [17]	DarkNet53	94.58	70.43	81	89.93	51.43	97.07	75.67	85	89.54	50.98	70	245MB	416×416
YOLOX [40]	DarkNet53	93.78	67.61	79	89.65	52.61	90.83	71.41	80	86.93	47.82	50	97MB	640×640
SSD300 [15]	VGG16	87.86	72.61	80	83.78	49.15	93.95	33.17	49	75.80	41.65	142	91MB	300×300
SSD512 [15]	VGG16	87.56	74.66	81	85.63	50.37	90.16	55.32	69	71.12	40.76	80	92MB	512×512
RetinaNet [18]	ResNet50	91.64	73.39	82	89.21	52.43	39.41	51.86	45	39.1	20.77	49	145MB	600×600
CenterNet [20]	ResNet50	94.71	60.16	74	89.03	52.36	92.66	59.7	73	79.82	44.69	127	124MB	512×512
Faster-CNN [14]	ResNet50	75.79	71.01	73	76.44	47.52	67.65	75.09	71	73.36	42.72	15	108MB	600×600
EfficientDet [41]	EfficientNet	89.61	71.88	80	86.35	49.41	94.42	39.86	56	69.83	40.35	45	15MB	512×512
YOLOV7 [42]	ELAN-CSPDarkNet	90.42	72.43	80	90.31	52.87	83.77	82.83	83	87.32	48.53	107	37MB	512×512
EfficientViT [43]	EfficientViT	88.68	75.04	81	90.06	51.94	88.38	81.63	85	90.08	50.38	86	61MB	512×512
TransNeXt [44]	TransNeXt	88.43	72.19	79	89.51	50.83	87.36	81.62	84	89.71	49.64	76	72MB	512×512
FocalNet-H [45]	FocalNet	89.14	74.61	81	90.64	52.72	89.82	79.39	84	88.53	50.37	69	107MB	512×512
3SD-Net + **EGS**	ResNet50	94.95	71.44	82	**91.66**	**53.95**	95.24	76.38	**85**	**90.22**	**52.79**	82	134MB	512×512
3SD-Net + **EDS**	ResNet50	**96.37**	61.79	75	91.58	53.44	**97.52**	74.71	**85**	89.73	52.55	82	134MB	512×512
3SD-Net + **LDS**	ResNet50	83.98	**85.15**	**85**	89.26	53.08	73.0	**88.82**	80	88.24	51.37	82	134MB	512×512



(a) (b) (c) (d) (e) (f) (g) (h) (i)

Fig. 13. Visual comparisons with other methods. (a) GT indicates the GT of the ship. (b) Ours means the results of our SARShipNet. The images are selected from the test set in the SAR-Ship dataset. Our 3SD-Net performs better at small object detection, while most methods fail in these cases. (a) GT. (b) 3SD-Net. (c) CenterNet. (d) YOLOV7. (e) EfficientDet. (f) EFFViT. (g) YOLOV4. (h) RetinaNet. (i) F-RCNN.

and APs by 4.78% in the SSDD dataset. In comparison to TransNeXt, our model increases precision by 7.37%, AP0.5 by 2.4%, and APs by 6.13% in the SAR-Ship dataset. Compared with FocalNet-H, our model increases precision by 6.03%, AP0.5 by 1.9%, and APs by 4.8% in the SSDD dataset. Compared with baseline CenterNet, our model does not significantly increase the model (↑ 10 MB) size and reduce the detection speed (↓ 45) but it increases precision by 2.58%, Recall by 16.68%, $F1$ by 12%, AP0.5 by 10.4%, and APs by 8.1%. Our 3SD-Net (*EGS*) model outperforms all other models on the SSDD dataset in terms of both AP0.5 and APs' metrics. While 3SD-Net (*EDS*) and 3SD-Net (*LDS*) variants show slightly weaker performance on AP0.5 compared with the top-performing model, they exhibit substantial improvements over YOLOv7, EfficientViT, TransNeXt, and FocalNet-H in APs.

2) Visual Comparisons: Fig. 13 shows a visual comparison of different methods. Our proposed 3SD-Net is much better than other methods for small object detection, which also shows the effectiveness of our proposed simple multiscale method MRF. We observe that SARShipNet can identify the majority of small objects, while other detectors ignore many smaller ships. Especially when the ships are gathered together,

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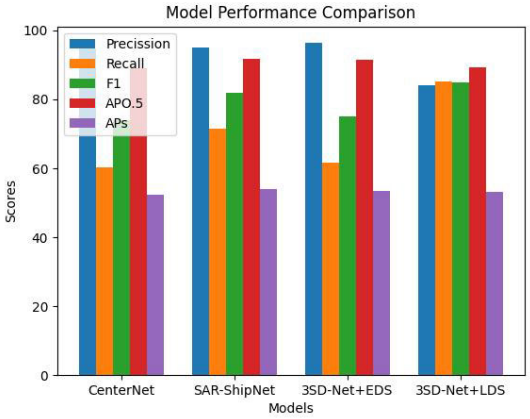


Fig. 14. Visual comparison results of models under different probability distributions. Among them, SAR-ShipNet is 3SD-Net combined with EGS. We can see that each probability distribution has different advantages.

3SD-Net can still accurately identify the majority of ships, while other detectors such as EfficientDet and Faster-RCNN miss many ships. By effectively solving the problems existing in SAR ship detection, the deep learning detector still achieves satisfactory detection results in this field.

Similarly, we also studied the influence of different probability distributions on the 3SD-Net model. The experimental results are shown in Table VII. Fig. 14 shows the visual comparison results of the 3SD-Net under different probability distributions. EGS probability distribution still achieves the best AP, which shows that elliptical Gaussian is very useful for ship detection. EDS and LDS perform better in other indicators. For example, EDS distribution achieves the best accuracy in the two datasets under the condition of significantly reducing AP, which is very suitable for task scenarios requiring high accuracy. LDS distribution can strike a balance between AP and recall rate. It also achieves a far higher recall rate than other detectors on two datasets, which may be very helpful for detection scenarios with high recall rates.

V. Conclusion

In this article, we propose a rapid and efficient framework for ship object detection, namely, 3SD-Net. The 3SD-Net effectively incorporates attention mechanisms, feature fusion, and center distribution while considering aspect ratios to accurately detect ship objects. This approach aims to address three key challenges in SAR ship detection: complex backgrounds, small targets, and ships with large aspect ratios. By effectively addressing the challenges inherent in SAR ship detection, the deep learning detector continues to yield satisfactory results. We conduct experiments on two popular ship datasets. Quantitative results show that our model is SOTA. Ablation experiments demonstrate the effectiveness of the proposed module. The detection pipeline of the entire 3SD-Net is simultaneously simplistic, yet achieves optimal detection results without compromising model size or inference speed.

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